Jonathan Tobias da Silva

Senior Embedded/R&D Engineer | Technical Leadership | DevOps

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Senior embedded and R&D engineer building reliable industrial and telecommunications systems, with technical leadership and DevOps experience.

Skills

Embedded & Platforms

Embedded Linux | Embedded Systems | Linux | Yocto Project

Languages

C/C++ | Python

Networking & Protocols

Networking | OPC UA | DLR | EtherNet/IP | L2 Protocols | L3 Protocols

Research

Industrial Automation | Cybersecurity

Leadership

Technical Leadership | Leadership

DevOps & Tooling

Git & CI/CD | DevOps

Experience

Senior Embedded Software Engineer

Act Digital

February 2026 – Present | Remote

- Develop and customize mission-critical embedded software for Embedded Linux and RTOS, focusing on industrial switches and PLCs.
- Strategic partnership with Schneider Electric on R&D projects for Industrial Automation and Industry 4.0.
- Implement industrial communication protocols, including EtherNet/IP, RSTP, DLR, OPC UA and Modbus.
- Optimize real-time system behavior and latency to meet the rigorous standards of Industrial IoT (IIoT) and critical infrastructure.

Embedded Software Engineer

Datacom

September 2024 – February 2026 | Remote

- Implemented Layer 3 protocols (IP, OSPF, BGP) and optimized system stability across hardware platforms.
- Developed and maintained software for high-performance telecommunications devices.
- Develop and debug embedded Linux systems, with a focus on performance tuning and system stability across various hardware platforms.
- Enhance code reliability through the design and execution of unit tests and automated integration tests.

R&D Engineer

Nova Smar

January 2021 – September 2024 | Sertãozinho, SP, Brazil

- Coordinate product development initiatives, managing a team of 10 engineers for embedded solutions.
- Apply cybersecurity measures for IEC 62443 compliance and O-PAS standards, enhancing the security and integrity of embedded systems.
- Design and implement embedded software for industrial automation, including PLCs and Windows Desktop applications.
- Develop a suite of internal Machine Learning automation tools in that reduced manual testing time and optimized engineering workflows.

Education

M.Sc. in Electrical Engineering

University of São Paulo (USP) | 2022 – 2025 | São Carlos, SP, Brazil

- Research on Industrial Automation Security and OPC UA protocol resilience against cyberattacks.
- Develop actionable security recommendations for protecting Industrial Automation and Control Systems (IACS) against evolving threats.

B.Sc. in Electrical Engineering

Federal Institute of São Paulo (IFSP) | 2017 – 2021 | Sertãozinho, SP, Brazil

- Specialize in Computer Science and Industrial Automation, completing multiple scientific research projects focused on Digitization and AI.

Technical degree in Administration

State Technical School (ETEC) | 2014 – 2016 | Batatais, SP, Brazil

- Foundational knowledge in business management and leadership.

Projects

UAnalyzer

Traffic analyzer for intrusion detection in industrial OPC UA networks.

GitHub

Forgenv

Creator and Lead Developer

A cross-platform dotfiles and environment manager with a beautiful TUI, designed for developers who want to manage their development environments with style and efficiency.

GitHub

Leetsheet

A TUI-based LeetCode cheatsheet for quick problem walkthroughs in your terminal.

GitHub

Publications

- Jonathan Tobias da Silva, A. L. Dias, I. N. da Silva. A Survey on OPC UA protocol: overview, challenges and opportunities. IEEE Xplore, December 2023. paper.
- Jonathan Tobias da Silva, O. Paschoal. Strengthening Industrial Automation: Best Practices in Cybersecurity. Controle & Instrumentação (Valete), May 2023. article.
- A. L. Dias, Jonathan Tobias da Silva, A. C. Turcato, G. S. Sestito. An intelligent fault diagnosis for centrifugal pumps based on electric current information. IEEE Xplore, August 2021. paper.